



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

ME1654-1 – INNOVATIONS AND DESIGN THINKING

(ME, EC, IS, AI&ML)

Duration: 2 Hours

Max. Marks: 50

Note: Answer Three full questions choosing One full question from each Unit

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain briefly the 5 different stages in the process of design thinking.	07	L1	1	1
	b) Explain 360° approach of design thinking with a suitable example.	08	L2	1	2
2.	a) Explain invention and innovation processes with examples.	05	L1	1	1
	b) How do you empathetically collect data from a person who is looking for a chair which he/she wants to use both for studying and relaxing purposes?	10	L2	1	2
Unit – II					
3.	a) Write the benefits of design thinking and business process modelling that are in common.	08	L1	2	1
	b) Explain the advantages of agile development method used in software development in IT industry.	07	L1	2	1
4.	a) Write a note on scenario based prototyping.	05	L1	2	1
	b) Explain in brief the role of agile management methodology in virtual collaboration.	10	L1	2	1
Unit – III					
5.	John is a vegetarian office going man whose office hours are between 9am to 6pm. He is looking for weight gain as he is lean. Moreover, he wants to be fit and hence he is regular to the gym also. He specifically likes white rice and salads to eat. He also prefers his food to be fresh and hot. Ideate a tiffin box that suits John's requirement wherein the cost of the tiffin box should not exceed 1500 rupees. Assume all necessary inputs.	20	L3	3	3
6.	There is a 1 acre farm land of 40X50 Sq ft. site. This land is very close to forest and gets all the serenity and tranquility of nearby forest with some wild animals habitat like forest giant squirrel, monkeys, deer, wild boar, snakes and some varieties of native birds. The land is placed in a downhill making the rain water flow downwards and pass through the channel located at the center of the land. A corporate working engineer is looking to build a farm house on this land. He has plans of having vegetables in one section of the land, some fruit garden in the other section, a small lake to cultivate some fishes in the third section which can in turn serve as the source of water for the land, fourth and fifth sections for dog kennels having 3 different breeds and a cow shed holding around 5 cows, respectively. Empathize with the engineer, define his requirements and ideate the farm land layout for his ease of operation and maintenance with necessary assumptions. He is ready to invest not more than 20 lakhs on this land. Also show the respective price allotment to each section.	20	L3	3	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
